

Science A–Z



Learning A–Z

level **U**

Grade

5–6

Lexile

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Unit • Mixing Matter

Multi-level

S T U

Word Count

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Nonfiction • Physical

KEY ELEMENTS USED IN THIS BOOK

The Big Idea: Everything is made of tiny particles called atoms. These atoms combine to form molecules. Atoms and molecules can combine with other atoms and molecules in many ways to form many different types of things. It is important to understand how materials change when combined. Some materials retain their own properties, while other materials form something new. Throughout our day, we see, use, and even consume combinations of materials. It is useful to understand how materials will react when combined. This knowledge can even keep us safe.

Key words: atom, chemical, chemical change, chemical reaction, concentrated, density, electron, element, gas, liquid, magnetism, mass, material, matter, mix, mixture, molecule, neutron, nucleus, physical change, property, proton, saturated, solid, solution, states of matter, substance, suspension, symbol, volume, weight

Key comprehension skill: Compare and contrast

Other suitable comprehension skills: Cause and effect; classify information; main idea and details; identify facts; elements of a genre; interpret graphs, charts, and diagrams

Key reading strategy: Visualize

Other suitable reading strategies: Ask and answer questions; summarize; connect to prior knowledge; using a table of contents and headings; using a glossary and boldfaced terms

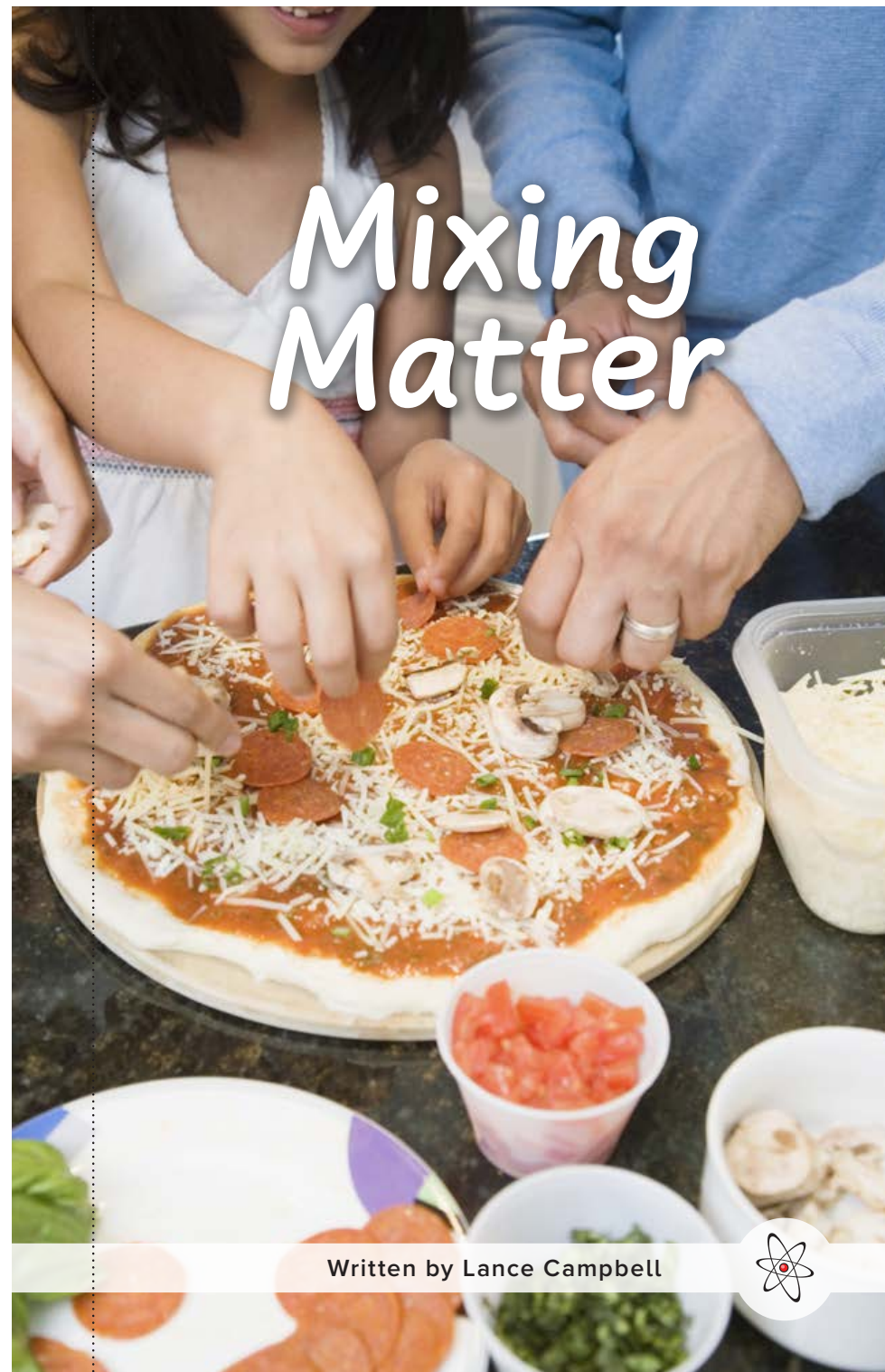
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Mixing Matter



Written by Lance Campbell



Mixing Matter

Written by Lance Campbell



Focus Question

What are the smallest particles that make up matter, and what are some ways that matter can combine?



Words to Know

atoms	matter
bond	mixtures
chemical change	molecule
chemical reaction	neutrons
compound	nucleus
concentrated	physical change
electrons	protons
elements	saturated
formula	solution
mass	suspension

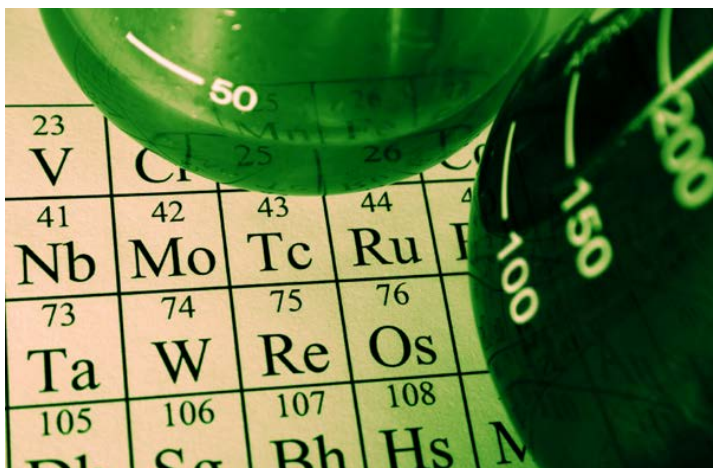


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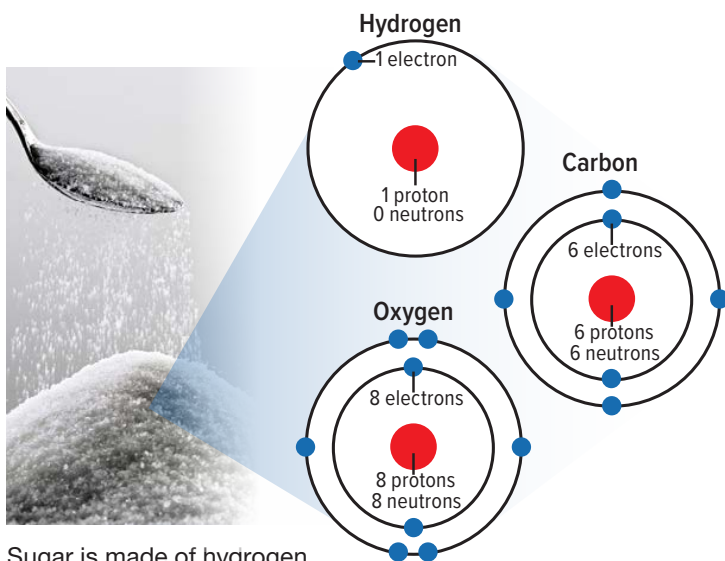
Introduction

Look at the things around you. You probably see things with many different properties. You may see books, a chair, and some water. All these things are types of matter. A balloon and the air you blow into it are types of matter. Everything in the universe that takes up space and has weight is made of matter. But what exactly is matter? And what makes the many kinds of matter different? How can matter be mixed to create new things? This book explores these and other questions to help you better understand what matter is and how it can change.

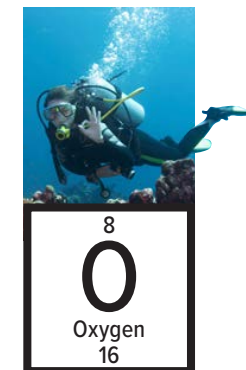
Elements

All matter is made of simple substances called elements. An element is a substance that cannot be broken down into a simpler substance. There are about 120 known elements on Earth.

Substances can be made of a single element. Pure iron, gold, and silver are each made of a single element. They cannot be made into simpler substances. But most substances we know of are a combination of two or more elements. Sugar is a substance made of three elements, while salt is a substance made of two elements. This means that both sugar and salt can be broken down into the simpler elements they are made of.



Sugar is made of hydrogen, carbon, and oxygen.



The number above the symbol shows how many protons are in an atom of that element. The bottom number is the average mass of a single atom of that element.

Each element can be identified using a symbol. Many symbols are the first one or two letters of the element's name. For example, helium is "He," carbon is "C," and oxygen is "O." The images above show some common elements and their symbols.

Word Wise

Some symbols for elements come from Latin.

Fe is the symbol for the element iron.

Fe is short for *ferrum*, the Latin word for iron. *Au* is the symbol for the element gold. *Au* is short for *aurum*, the Latin word for gold.



iron ore

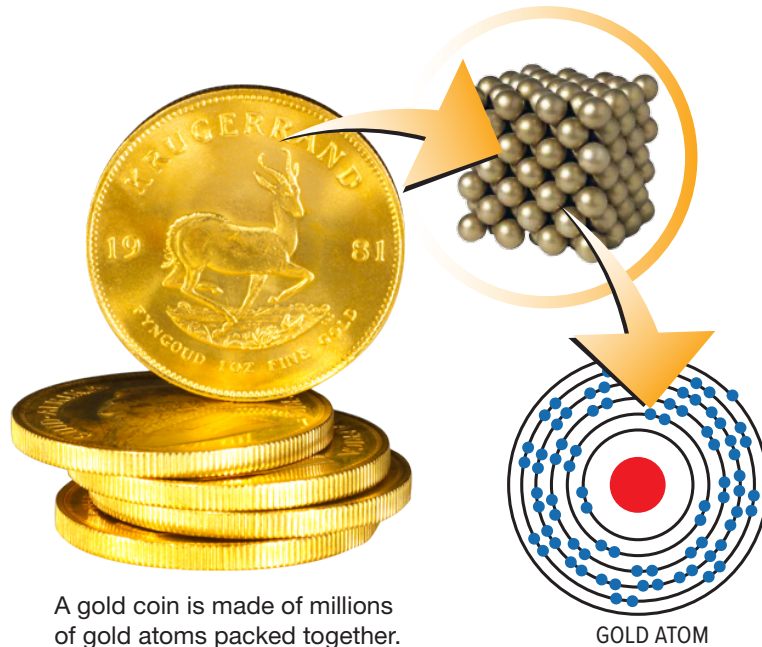


gold

Atoms

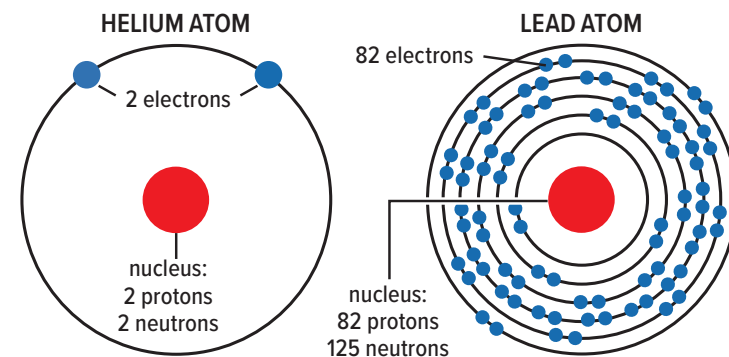
Imagine that you had the power to look deep inside any substance or material. If you had this power, you would see that all matter, and therefore all elements, are made of tiny, invisible particles called **atoms**. Different elements are made of different atoms.

Atoms are the smallest part of an element. If you could look into a gold coin, you would see that it is made of millions of gold atoms. If you could see inside a balloon filled with helium, you would see that the helium gas is made of millions of helium atoms.

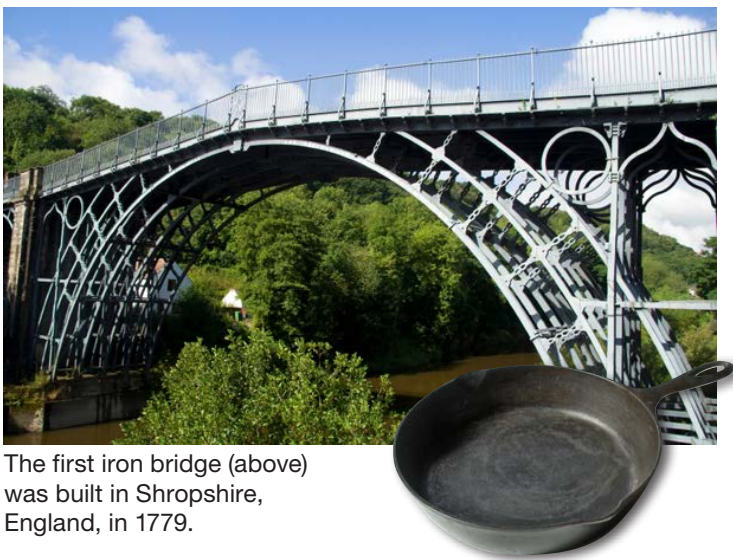


Let's further imagine that you had super powers that let you see inside an atom. You would see that atoms are made of even smaller particles. The structure of an atom is in some ways like the structure of the solar system. In the solar system, planets orbit, or go around, the Sun. In an atom, tiny particles called **electrons** orbit the **nucleus**, or center of the atom. The nucleus of an atom is like the Sun around which planets revolve. The nucleus is made of two kinds of particles—**protons** and **neutrons**. All atoms of the same element have the same number of protons and electrons.

As tiny as atoms are, they have **mass**. Most of the mass of an atom is in the nucleus. So the more protons and neutrons an atom has, the more mass it has. A helium atom has 2 protons and 2 neutrons in its nucleus. This makes helium the second-lightest element. (The lightest is hydrogen.) Lead, on the other hand, has 82 protons and 125 neutrons, so it is much heavier than helium.



The more atoms you have in a sample of an element, the more it will weigh. For example, an iron skillet weighs less than an iron bridge because there are billions and billions more atoms in the bridge than in the skillet.



The first iron bridge (above) was built in Shropshire, England, in 1779.

Do You Know?

In ancient Greece, people thought that matter was made of just four elements: earth, air, fire, and water. People believed this idea for thousands of years. It was not until the 1600s that people began to realize that there are many elements. None of those elements are earth, air, fire, or water.

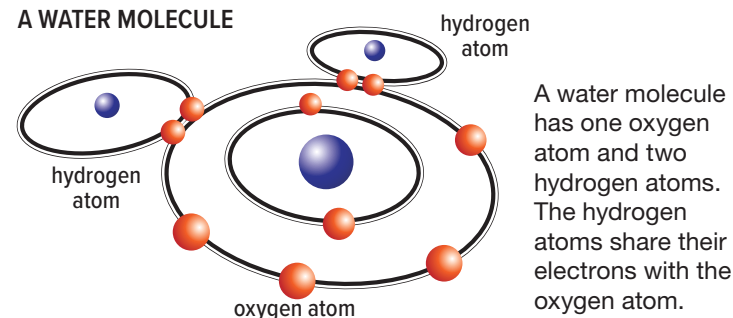
Combining Matter

You just learned that there are about 120 elements and that each is made of a unique kind of atom. But how can just 120 or so elements make up the millions of materials and substances found on Earth? They can because atoms combine.

Sometimes when atoms get close together and conditions are right, they start to share electrons. Atoms that share electrons form a **bond**. When atoms form or break a bond, they produce a **chemical reaction**. The result of atoms forming bonds is a new and bigger particle called a **molecule**.

Molecules can be a combination of just two atoms or many more, and those atoms can be of the same element or different elements. For example, a molecule of water has 3 atoms, while a molecule of table sugar has 45 atoms. Even though molecules can be made of dozens of atoms, they are still too tiny to see.

A WATER MOLECULE



A water molecule has one oxygen atom and two hydrogen atoms. The hydrogen atoms share their electrons with the oxygen atom.

When two or more different elements join, they make a **compound**. Each compound has its own properties. Salt has properties that are different from sugar because it has a different kind of molecule. A molecule is the smallest particle of a compound.

Compounds also have properties that are different from the elements that make them up. Water is a compound as well as a molecule. Oxygen and hydrogen are commonly found in the form of *gases*. But when they combine chemically, they form *liquid* water.

Molecules and compounds, like atoms and elements, are identified using letters and numbers. The letters show the elements that make up the compound, and the numbers represent the number of atoms of each element in a molecule of the compound. Common table sugar, for example, has 12 carbon atoms, 22 hydrogen atoms, and 11 oxygen atoms. Therefore, its **formula** is $C_{12}H_{22}O_{11}$.

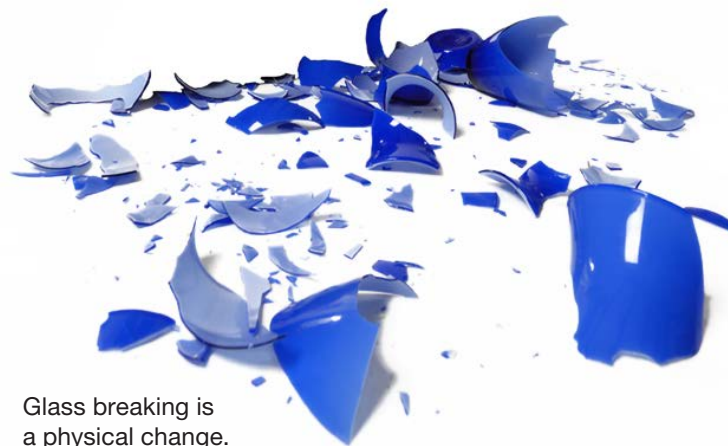


Rock candy is simply colored table sugar with large crystals.

Changing Matter

When substances are combined, they can simply mix together without bonding to make a new compound. If the substances do not form a totally new substance, any change you observe is called a **physical change**. But when a new substance forms from the mixing of two or more different substances, the change in substances is a **chemical change**.

If matter changes from one state to another, such as liquid water changing to solid ice, it is a type of physical change. Evaporation, which changes liquid water into water vapor, is also a physical change. In each case, whether a solid, liquid, or gas, it is still water. Breaking glass into many pieces is another type of physical change. In each of these examples, no new substance is formed.



Glass breaking is a physical change.

Iron oxide, or rust, is used in paint, ink, rubber, plastic, ceramic, brick, paper, glues, and many other products.



When sugar is mixed with water, a physical change occurs, and you get sugar water. The sugar and water did not change to make a totally new substance with different properties. The substance is still water—it just tastes sugary. But when oxygen in the air combines with iron, a new substance, called *rust* or *ferric oxide*, forms. Rust is not at all like oxygen or iron. It is an entirely new substance with its own properties. The formation of rust is a chemical change.



Unpainted iron rusts more quickly than painted iron.

There are five main ways to tell whether a chemical change is happening:

1. The substance or mix of substances *changes color*. Apples turn brown due to a chemical reaction with the air.



2. *Energy gets absorbed or released*. Many fireworks make light, heat, and sound when they burn.



3. The substance *changes odor*. Food smells bad when it spoils because of chemical changes.



4. The substance or mix of substances *produces gases or solids*. If you drop an antacid tablet into water, it will create gas bubbles in the water.



5. The change is *not easy to undo*. If you bake a cake, it would be very difficult to separate out its ingredients again because they've undergone chemical changes.



When you bake a cake, you might smell the cake baking and see the batter rise and turn brown. When the cake is done and sliced, you might see a spongy texture created by the gas bubbles that formed in the batter.



Cake batter is a simple mixture. But once it is baked, chemical changes happen.

Mixing Matter

Now let's take a closer look at the physical changes that occur when matter is mixed together. Different states of matter can be mixed together without resulting in a chemical change. You can mix solids with solids, solids with liquids, liquids with liquids, gases with gases, and liquids with gases. Remember, these are the changes that happen when elements or compounds simply mix together without forming new compounds that have new and different properties. These combinations of substances are called **mixtures**.

Sometimes it is easy to see the different materials in a mixture. At other times, different substances mix so well that it is hard to see the different parts that were mixed together. For



example, if you mix sand with water in a jar and shake it up, you just get cloudy-looking water. If you let the jar sit for a short time, the sand will easily separate and settle to the bottom of the jar.

Mixing Solids and Liquids

If you mix sugar or salt with water and shake it up, the salt or sugar dissolves and will not settle to the bottom if you let it sit. This kind of mixture is called a **solution**.

Adding different amounts of a solid to a liquid can make different strengths of solutions. For example, to make a stronger cup of hot chocolate, you can just add more chocolate powder. We say that the hot chocolate is more **concentrated** when it has more chocolate powder.



However, there is a limit to the amount of solid you can add to a liquid when making a solution. For example, if you add too much salt to water and shake it up, some of the salt will not dissolve and will just settle to the bottom. If a liquid solution cannot hold any more solid, it is called **saturated**. You can add more salt to a saturated solution of salt water by heating the solution.

COMMON MIXTURES, SOLUTIONS, AND SUSPENSIONS		
Substance	States of Matter	Mixture, Solution, or Suspension
salt water 	solid and liquid	solution
soda pop 	liquid and gas	solution
tossed salad 	solids	mixture
gelatin 	solid and liquid	suspension

There are many ways to distinguish the different substances in a mixture. One way is to look at their physical properties, such as volume, weight, density, and magnetism.

SOME PROPERTIES OF MATTER		
Property	Definition	Discussion
Volume	a measure of how much space something fills	Solids, liquids, and gases all take up space. It is impossible for one grain of sand to occupy the same space as another grain of sand.
Weight	a measure of how much mass something has	All matter has weight. Even a gas such as air has weight.
Density	a measure of how heavy something is compared to the volume it takes up	Almost all rocks are denser than water, so they sink when dropped into water. Most wood is less dense than water, so it floats in water.
Magnetism	the ability to be attracted by a magnet	Some matter, such as iron, is easily magnetized. Other types of matter, such as aluminum, wood, and paper, are not.

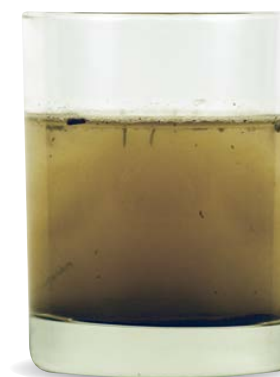


Cream is a special kind of suspension. Pieces of proteins and fat are suspended in water. If you shake the cream long enough, the fat droplets collect into a lumpy solid called butter!



Another mixture of solids and liquids is called a **suspension**. In a suspension, the pieces of solid material are so tiny and light that they do not immediately settle out. You cannot see the solid particles. This is a physical change because the solid materials do not change chemically.

If you put a spoonful of dirt into a glass of water, the tiniest particles form a suspension. You do not see dirt. You do not see clear water. You see brown water. If it is not disturbed, the tiny particles of dirt eventually settle to the bottom of the glass due to the force of gravity.

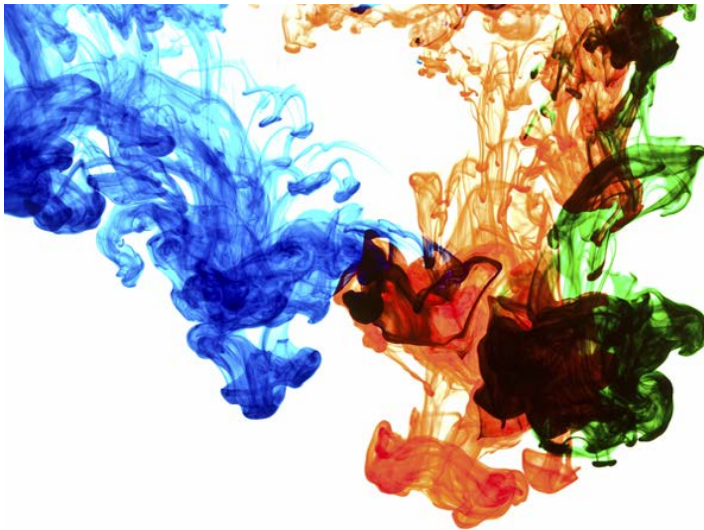


Mixing Liquids

Liquids can mix in different ways, too. Sometimes when you mix two liquids together, they stay separate. It is easy to see the separate liquids. If you mix oil and water, the oil floats on top of the water.



But in other liquid mixtures, the liquids do not stay separate. If you mix food coloring and water, the two liquids blend completely. It is difficult to separate them.



Conclusion

Everything we see, touch, and smell is matter, and all matter is made from a small group of elements. These elements combine in many ways, both chemically and physically, to make all the wonderful things you see around you. Without elements and the many combinations they make, our world could not exist.



Glossary

atoms page 7

the smallest parts of an element

bond page 10

a connection between atoms that are joined together to form a molecule

chemical change page 12

a change in the chemical makeup of a substance

chemical reaction page 10

a process in which one substance is changed to another

compound page 11

a combination of two or more elements

concentrated page 17

how much of a substance is present in a solution

electrons page 8

particles that are part of an atom and that orbit the nucleus; they have a negative electrical charge

elements page 5

substances that cannot be broken down into simpler substances

formula page 11

numbers and letters that describe how many atoms of each element are included in one molecule or compound

mass page 8

the measure of the amount of matter in an object

matter page 4

anything that takes up space and has weight

mixtures page 16

combinations of substances in which chemical reactions do not occur

molecule page 10

the smallest part of a substance that can exist by itself, made of two or more atoms

neutrons page 8

particles in the nucleus of an atom that have no electrical charge

nucleus page 8

the positively charged central region of an atom, consisting of protons and neutrons, and containing most of the atom's mass

physical change page 12

a change in the size, shape, or color of a substance that does not change it into a different substance

protons page 8

tiny particles that are part of the nucleus of an atom; they have a positive electrical charge

saturated page 17

being at the point at which no more of a liquid, solid, or gas can be absorbed by a solution at a given temperature

solution page 17

a mixture in which the atoms of a solid separate and become invisible in a liquid

suspension page 20

a mixture of a liquid and a solid in which the solid does not dissolve

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